

English Abstract

A Two-Layer Defense Architecture Against Suicide USV Swarms: Score-Map Cooperative Reinforcement Learning with a Large Language Model Commander, and Its Quantitative Validation

Suicide unmanned surface vehicle (USV) swarms resist conventional counters: electronic warfare cannot reach wire-guided targets, and interception spends a missile costlier than its target. We adopt physical capture: defense boats slower than the attackers pre-block the mothership approach corridor with net walls. Two layers update on the *same* decision period but differ sharply in *response latency*, so only the slow layer is decoupled asynchronously. The maneuver layer rasterizes the battlefield into a mothership-centered 15 channel grid; a FiLM-conditioned U-Net emits a score map; each boat points at two high-scoring pixels in its valid region and lays a net wall between them. Pixel pointing keeps the action space grid-regularized and the input invariant to swarm size. Training extends critic-free group relative policy optimization to the multi-agent case, with counterfactual credit assignment separating individual contribution. In the strategy layer a large language model (LLM) commander assigns boats to enemy clusters from pre-computed geometry, with no combinatorial post-optimization.

We collected 720 seed-paired episodes: 8 conditions $\times$ 3 formations $\times$ 30 seeds. The maneuver layer alone raised capture rate from 0.844 to 0.886 (+0.157 under wave), while the strategy layer split by model — a cloud model gained +0.060, whereas Qwen2.5 7B was significantly *worse* than the heuristic at -0.192 . In a 2×2 decomposition over 10 metrics, 9 interaction intervals contained 0: the layers are *additive*, not synergistic. Under Bonferroni correction capture rate and breaches lose significance; the other 8 metrics hold. The capture-rate effect size ($d_z = 0.44$) ranked lowest of the 10, far below total turning (-3.23) and nets per capture (-2.09). Commander skill tracked plan stability across replans, not assignment coverage.

Keywords : unmanned surface vehicle; swarm defense; multi-agent reinforcement learning; task assignment; large language model; semantic segmentation; group relative policy optimization

Highlights

- Two judgment layers share one 25 s period; only the slow one is decoupled
- Pointing at score-map pixels keeps the action space invariant to swarm size
- Critic-free group relative optimization with counterfactual credit assignment
- Across 720 paired episodes the two layers are additive in 9 of 10 metrics
- Commander skill is explained by plan stability, not assignment coverage